

Notes

Assignments

7.21, 7.23

Tutorial problems

7.25, 7.37, 7.40

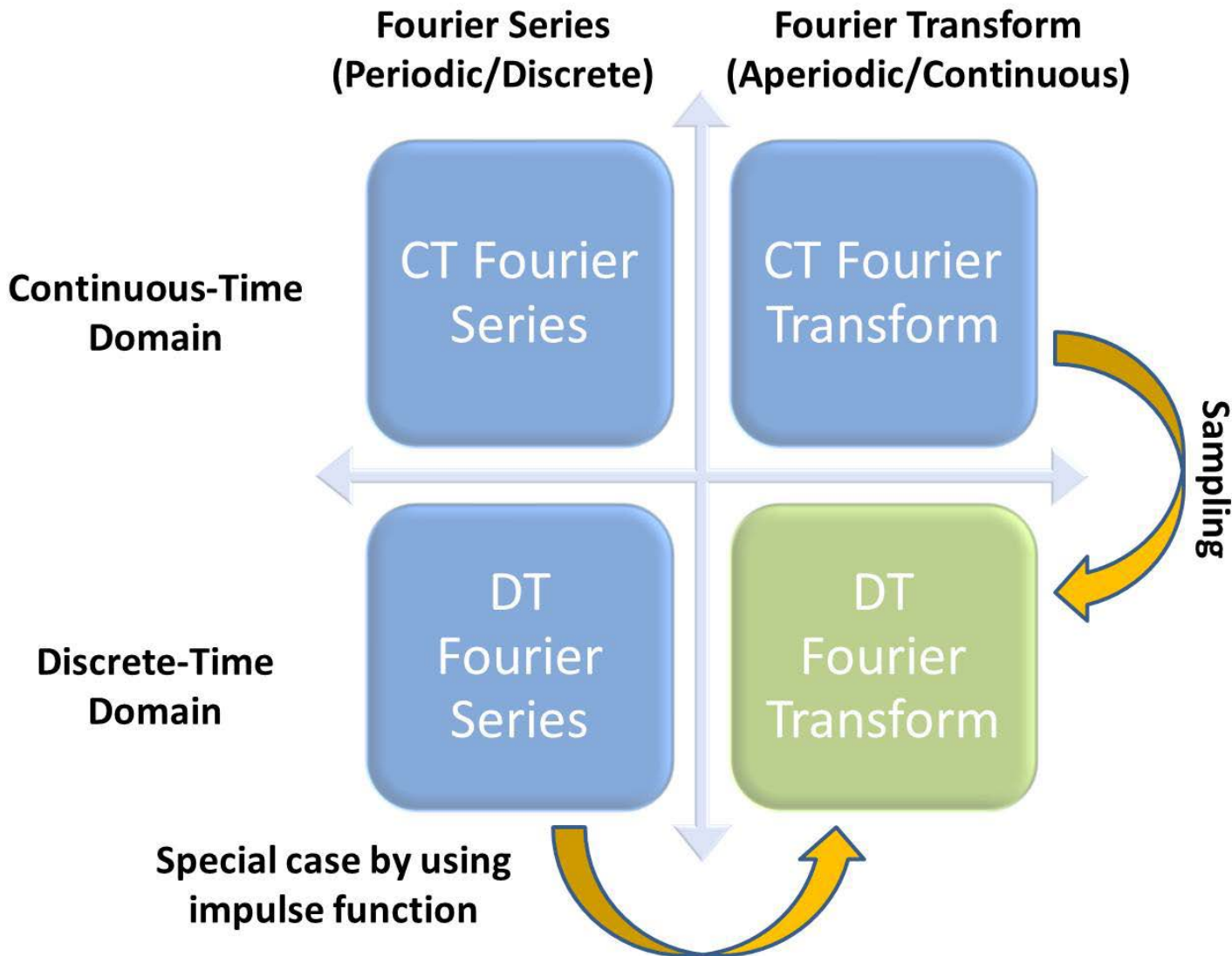


南方科技大学 SUSTC

Chapter 7: Sampling

Frequency domain

Time domain



CTFT Properties: Multiplication Property

thus if

$$x(t) * y(t) \longleftrightarrow X(j\omega) \cdot Y(j\omega)$$

then the other way
around is also true

$$\begin{aligned} x(t) \cdot y(t) &\longleftrightarrow \frac{1}{2\pi} X(j\omega) * Y(j\omega) \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\theta) Y(j(\omega - \theta)) d\theta \end{aligned}$$

— A consequence of *Duality*

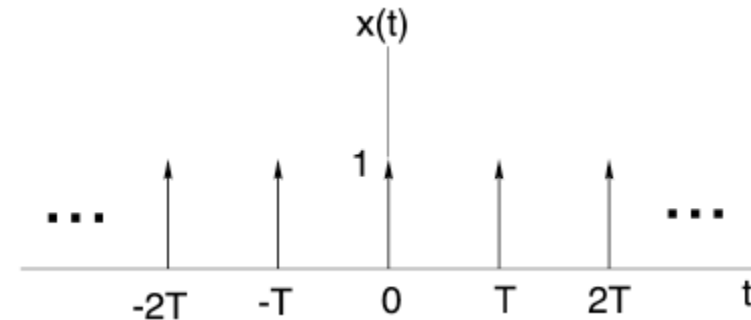
Example 4.8

$$x(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT) \quad \text{— sampling function}$$

$$x(t) \xleftrightarrow{\text{FS}} a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_o t} dt = \frac{1}{T}$$

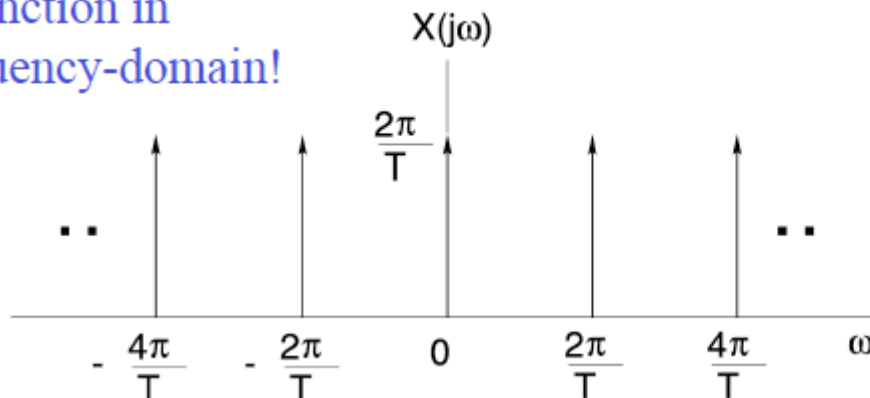
$$\Downarrow x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_o t} = \sum_{k=-\infty}^{+\infty} \frac{1}{T} e^{jk\omega_o t}$$

$$X(j\omega) = \sum_{k=-\infty}^{+\infty} \frac{2\pi}{T} \delta\left(\omega - \underbrace{\frac{k2\pi}{T}}_{k\omega_o}\right)$$



$\omega_s = 2\pi / T$: sampling frequency

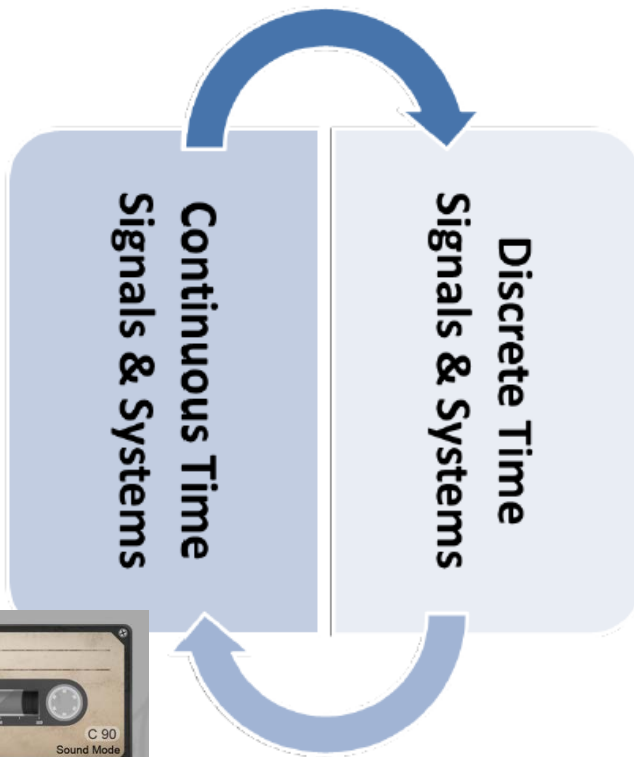
Same function in
the frequency-domain!



Note in this case, periodic
in both time domain (with
a period T) and frequency
domain (with a period
 $2\pi/T$)

Introduction

Sampling: to facilitate digital processing via computers or chips



Any lossless conversion?

Process CT signals with DT systems?



Interpolation: to present the output of digital processing

Example: Video recording

- **Signal to be sampled:** real scene (continuous-time signals)
- **Sampling:** record by camera with a rate of 24, 25 or 30 frames per second
- **Sampled signal:** video tapes, mp4 files, avi files and etc. (discrete-time signals)
- **Reconstruction:** watch via eyes and interpret in the brain
- In our consciousness, the real scene can be reconstructed without information loss



Outline

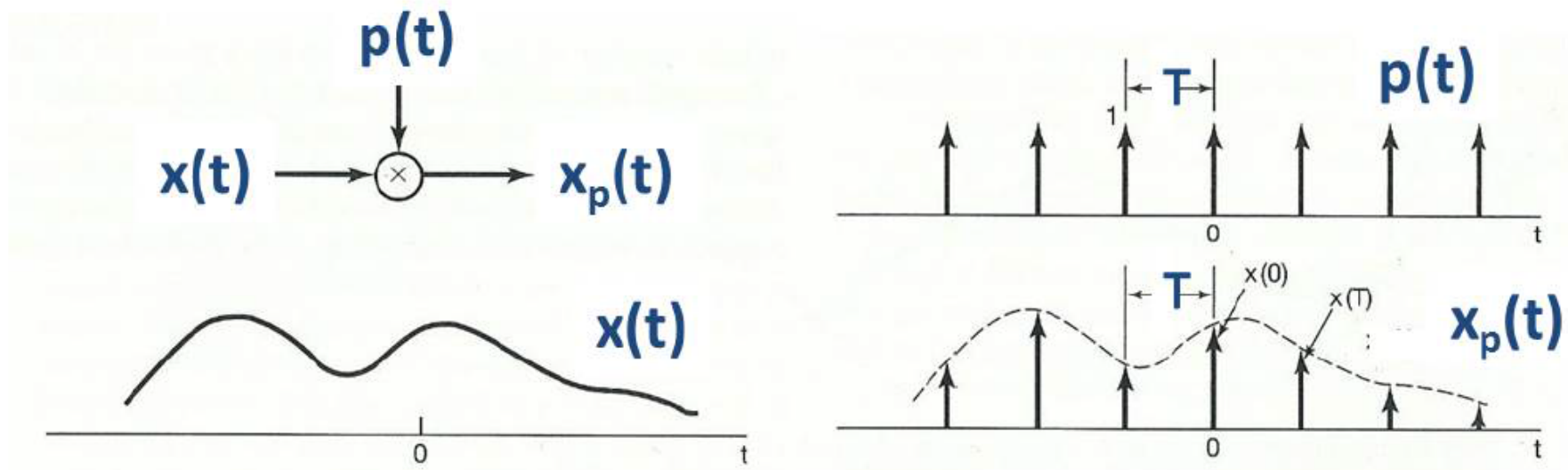
- Sampling is a general procedure to generate **DT signals** from **CT signals**, where information of the original signals can be kept
- **Core sampling theory:**
 - ◆ Impulse train, zero-order hold, 1st-order hold, etc.
 - ◆ Analysis in frequency domain
 - ◆ Nyquist rate
- **Undersampling:** Aliasing
- **Application:** process continuous-time signals discretely
- **More sampling techniques:** decimation, downsampling and upsampling

Note on digital signal

- Need be both discrete time (DT) and digital value
 - ◆ By sampling: sampling rate
 - ◆ By quantization: how many bits
 - ◆ Could be implemented by analogy-to-digital (A/D) converter

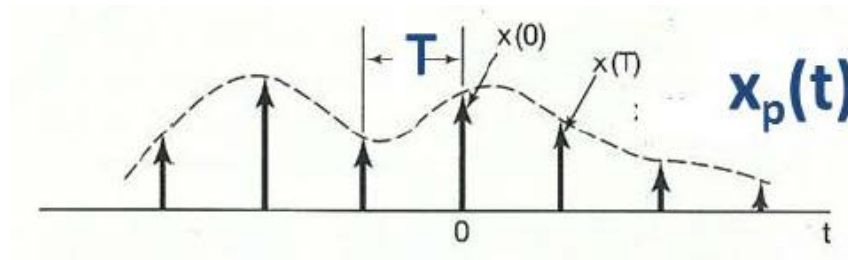
Impulse-train sampling

- Mathematically, sampling can be represented by multiplication



- Sampling function: $p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$
- Sampling period: T
- Sampling:

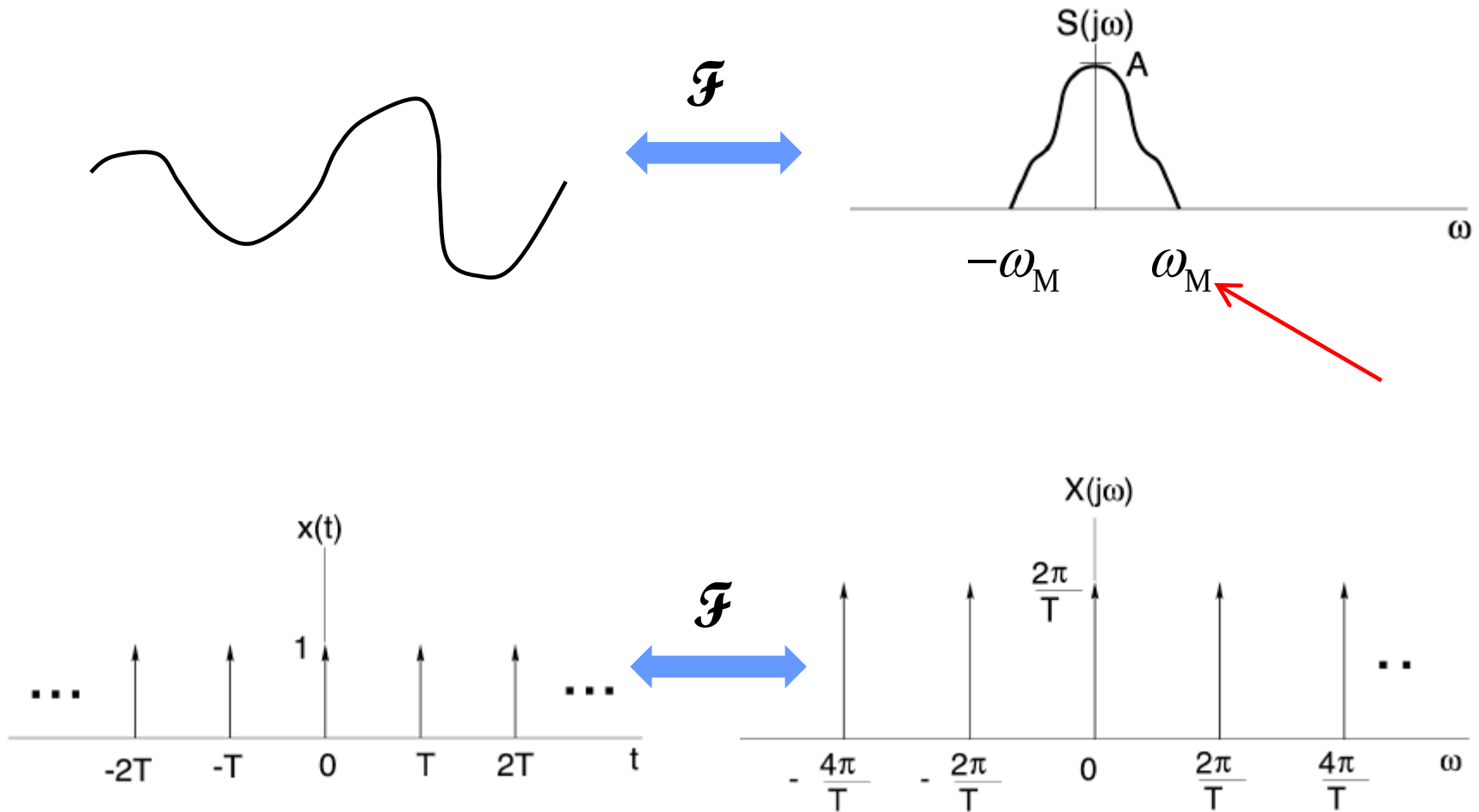
$$x_p(t) = x(t) \times p(t) = x(t) \times \sum_{n=-\infty}^{\infty} \delta(t - nT) = \sum_{n=-\infty}^{\infty} x(nT) \delta(t - nT)$$



Sampling discards most of points in the original signals.

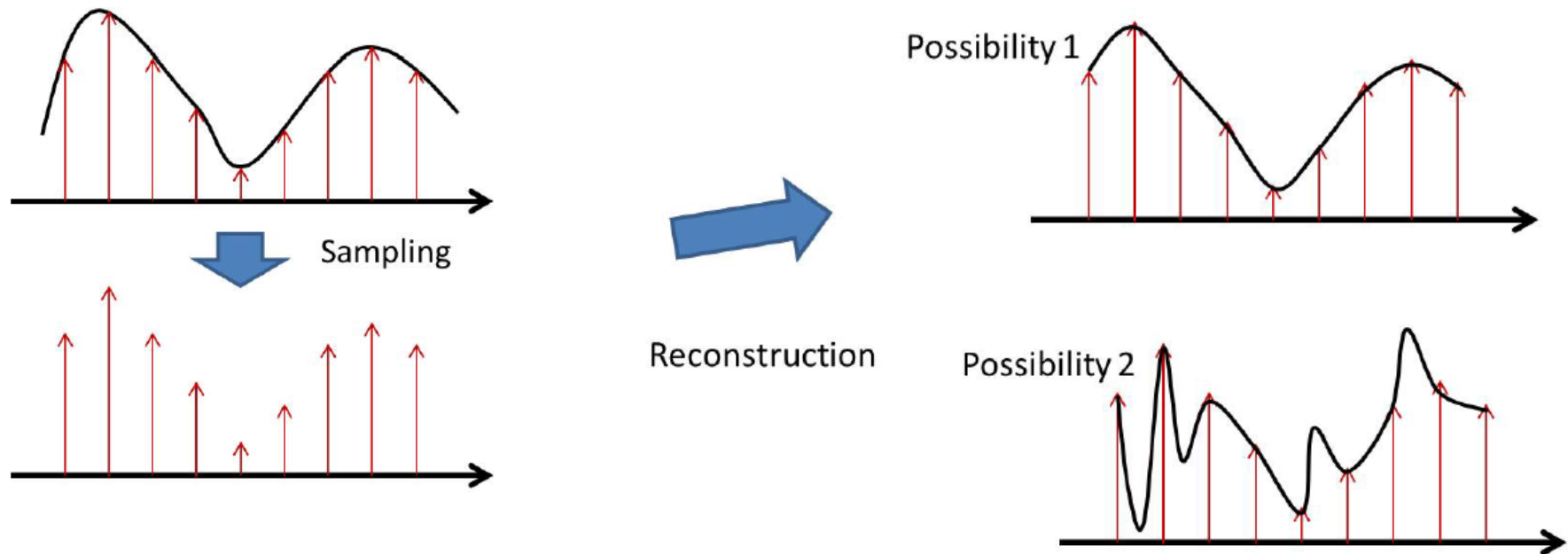
Is there any information loss in sampling?
Or can we perfectly reconstruct the original signal?

Two important frequencies



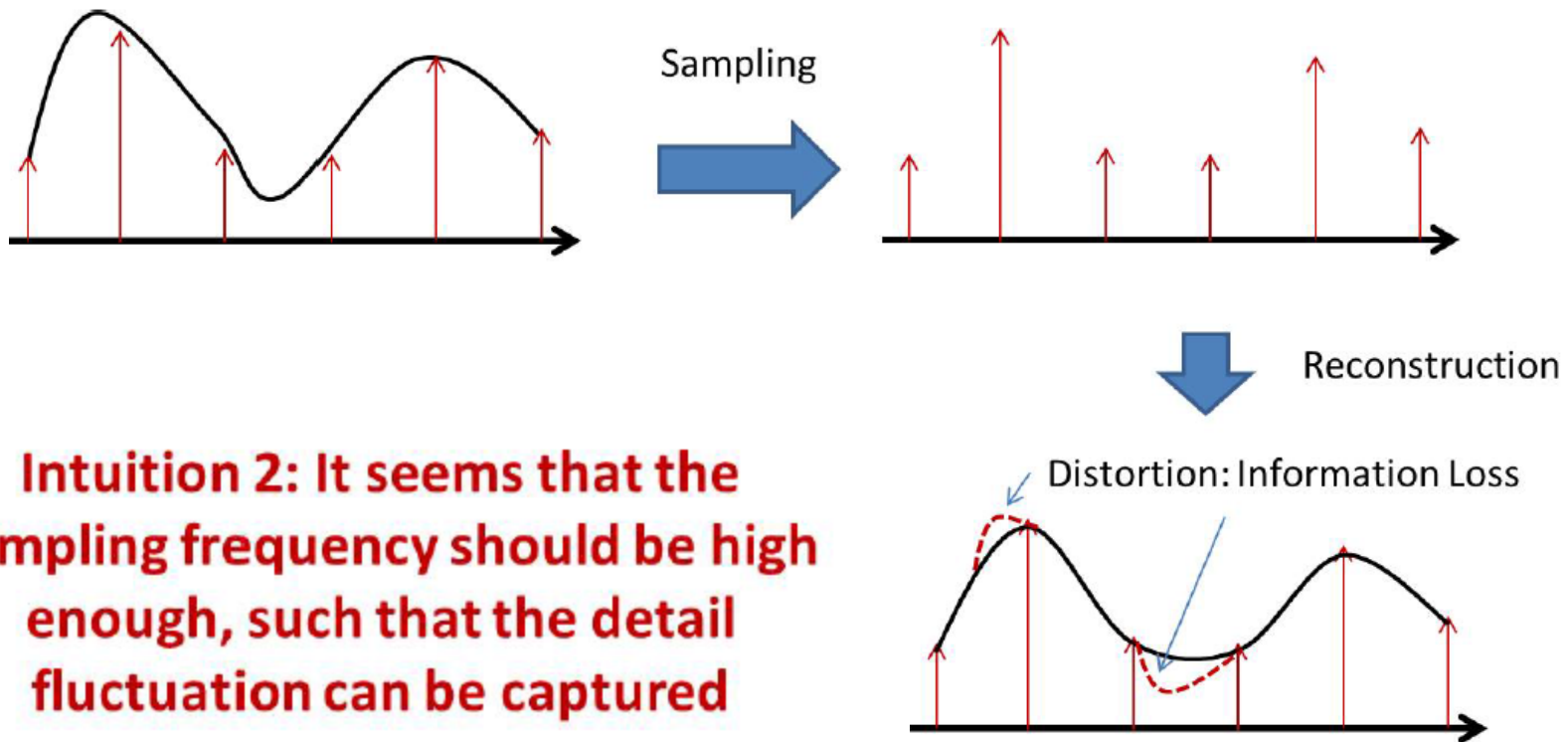
$$\omega_s = \frac{2\pi}{T}$$

Observation (1/2)



Intuition 1: It seems that we need a smooth interpolation

Observation (2/2)



- Sampling: the frequency should be high enough
- Reconstruction: the interpolation should be smooth enough

Frequency analysis (1/2)

- Theoretical tool: continuous-time Fourier transform
- Principle:

$$x(t) \times p(t) \Leftrightarrow \frac{1}{2\pi} X(j\omega) * P(j\omega)$$

- Fourier series of $p(t)$:

$$\begin{aligned} a_k &= \frac{1}{T} \int_{-T/2}^{T/2} \sum_{n=-\infty}^{\infty} \delta(t - nT) e^{-jk\omega_s t} dt \quad \text{where} \quad \omega_s = \frac{2\pi}{T} \\ &= \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_s t} dt = \frac{1}{T} \end{aligned}$$

- Fourier Transform of $p(t)$:

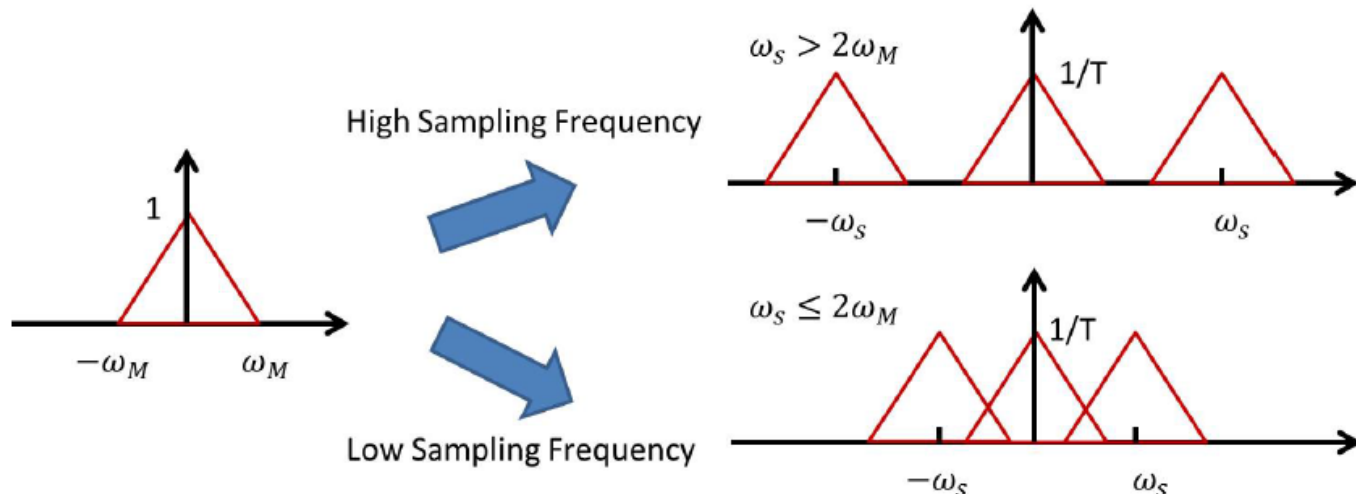
$$P(j\omega) = 2\pi a_k \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s) = \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s)$$

Frequency analysis (2/2)

- Fourier transform of sampled signal $x_p(t)$:

$$\begin{aligned} X_p(j\omega) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\theta) P(j(\omega - \theta)) d\theta \\ &= \frac{1}{T} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s)) \end{aligned}$$

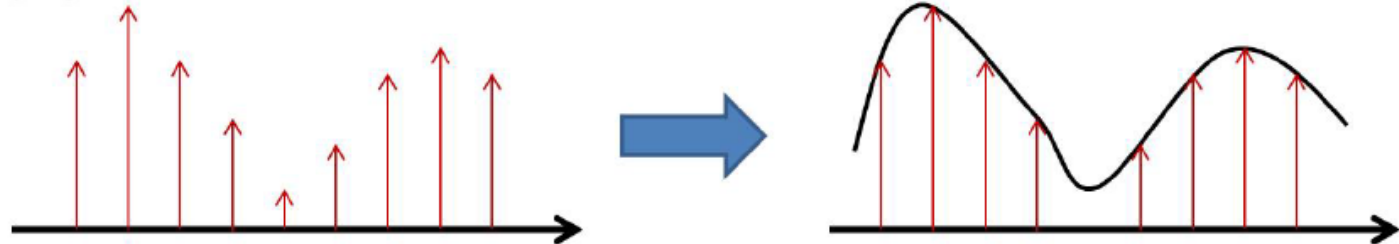
- Sampling: the Fourier transform of input signal is repeated with period ω_s



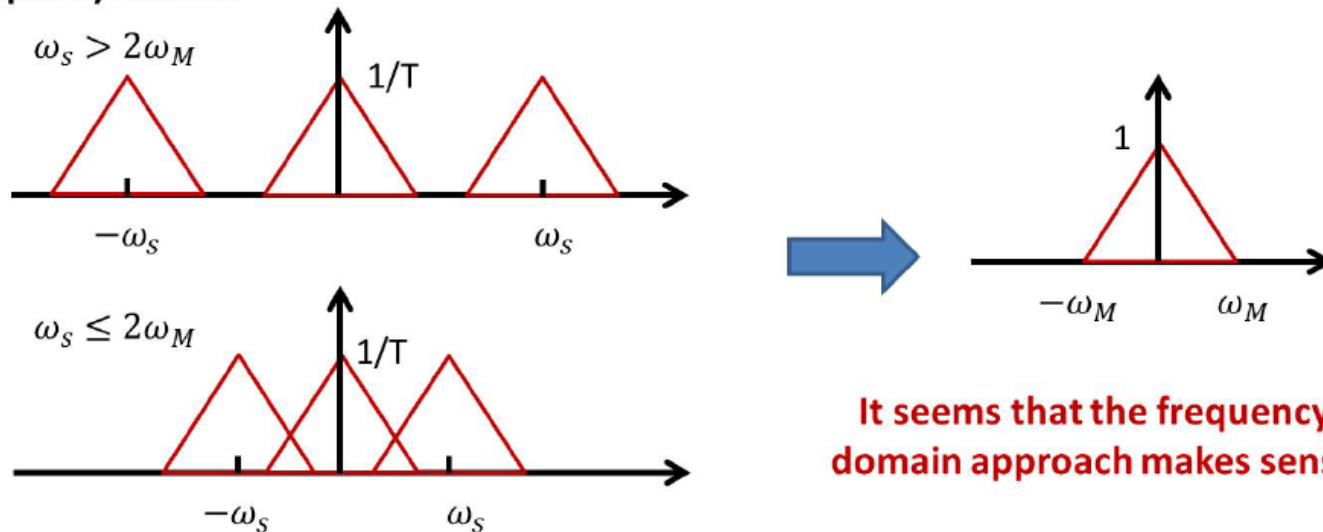
Reconstruction problem

- Given the sampled signal, can we perfectly reconstruct the signal before sampling?

Time Domain



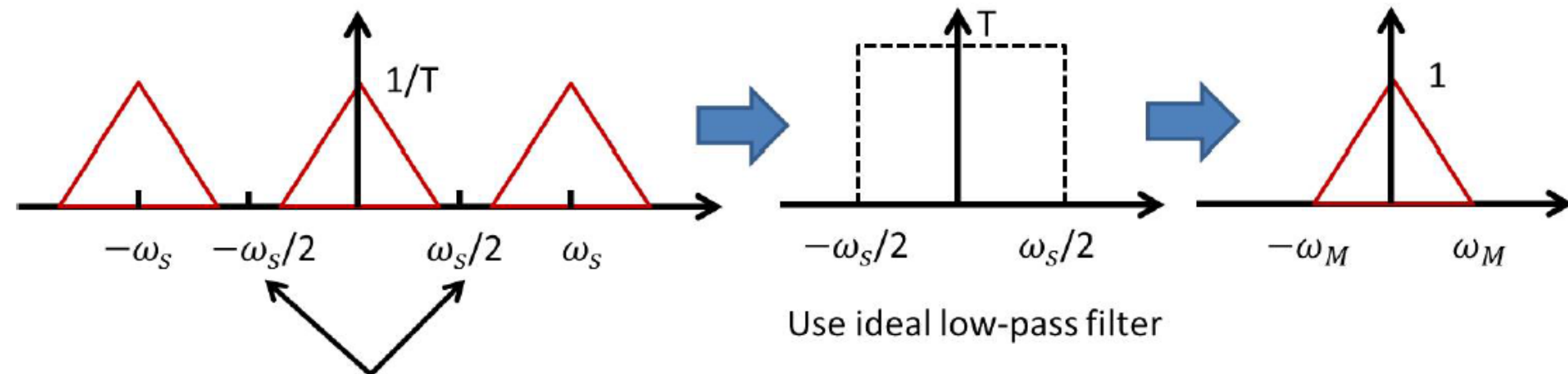
Frequency Domain



It seems that the frequency domain approach makes sense

Reconstruction (1/2)

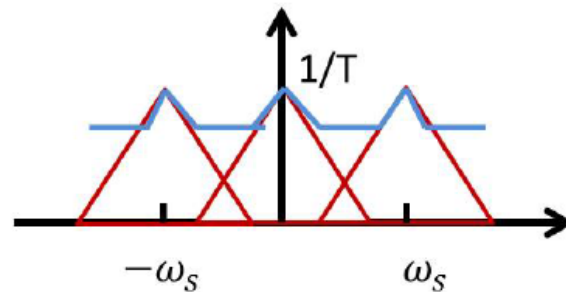
- Scenario of $\omega_s > 2\omega_M$



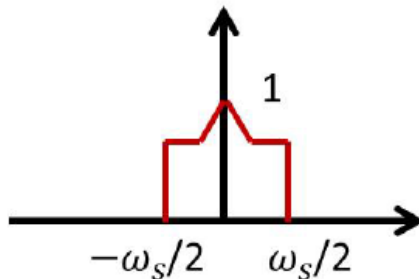
The spectrum of desired signal is within $\left(\frac{-\omega_s}{2}, \frac{\omega_s}{2}\right) \rightarrow$ No overlapping

Reconstruction (2/2)

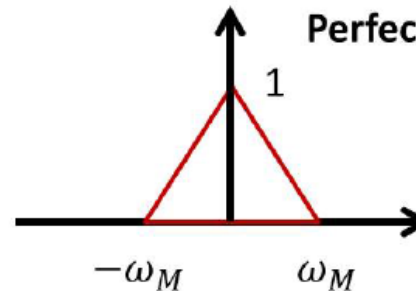
- Scenario of $\omega_s \leq 2\omega_M$



Since $\omega_s \leq 2\omega_M$, we don't know the frequency range of the desired signal
Sampling on the following signals can generate the same result:



OR



Perfect reconstruction is impossible

Observation: the original signal $x(t)$ can be Uniquely and perfectly reconstructed from $x(nT)$ only when $\omega_s > 2\omega_M$

Sampling theorem

Sampling Theorem

Let $x(t)$ be a band-limited signal with

$$X(j\omega) = 0 \text{ for } |\omega| > \omega_M.$$

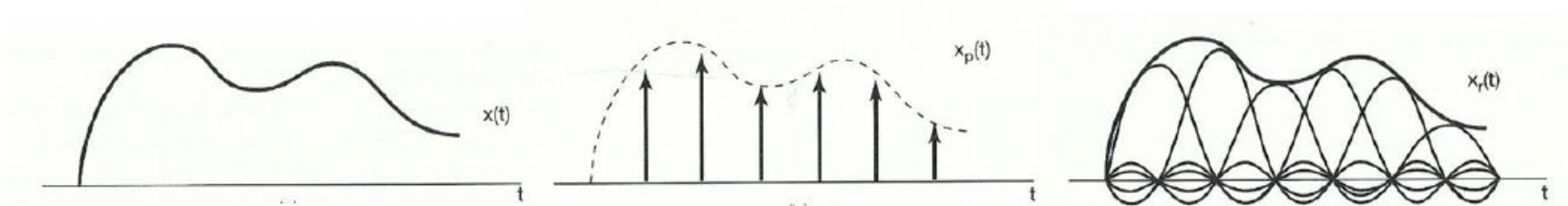
Then, $x(t)$ is uniquely determined by its samples $x(nT)$ or $x_p(t)$ if

$$\omega_s = \frac{2\pi}{T} > 2\omega_M,$$

where $2\omega_M$ is referred to as the *Nyquist rate*.

Signal reconstruction: Interpolation

- If $\omega_s > 2\omega_M$, original signal can be perfectly reconstructed by ideal low-pass filter.
- Time domain interpretation of lowpass filtering



$$\begin{aligned}
 x_r(t) &= x_p(t) * h(t) = \sum_{n=-\infty}^{+\infty} x(nT)h(t - nT) \\
 &= \sum_{n=-\infty}^{+\infty} x(nT) \frac{\sin \frac{\omega_s}{2}(t - nT)}{\frac{\omega_s}{2}(t - nT)} = \sum_{n=-\infty}^{+\infty} x(nT) \text{sinc}\left(\frac{t - nT}{T}\right)
 \end{aligned}$$

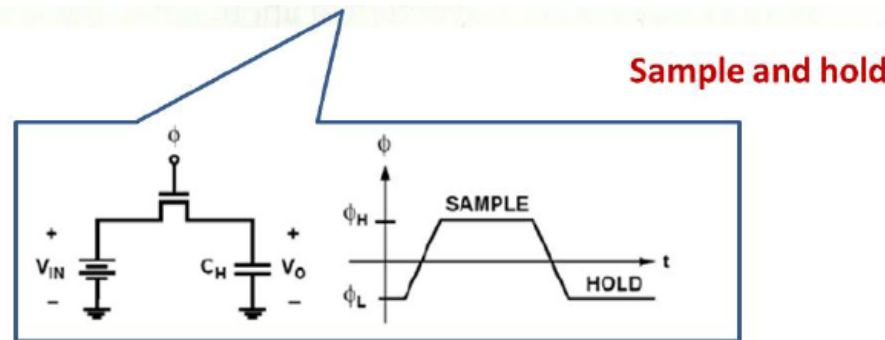
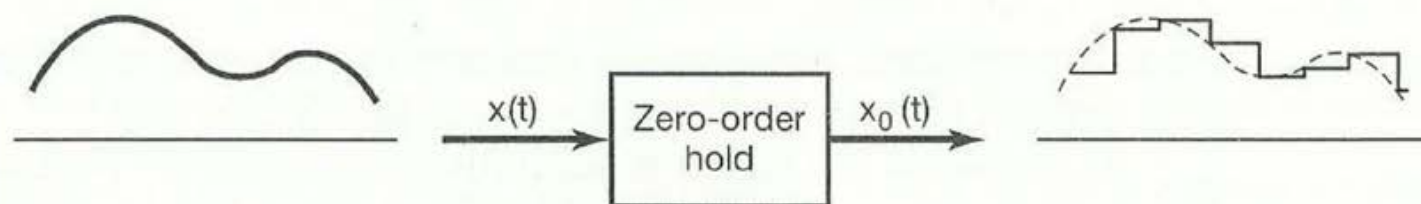
- Ideal lowpass filtering: interpolation with sinc function

Zero-order hold

- It's difficult to generate ideal impulse chain in practical implementation.

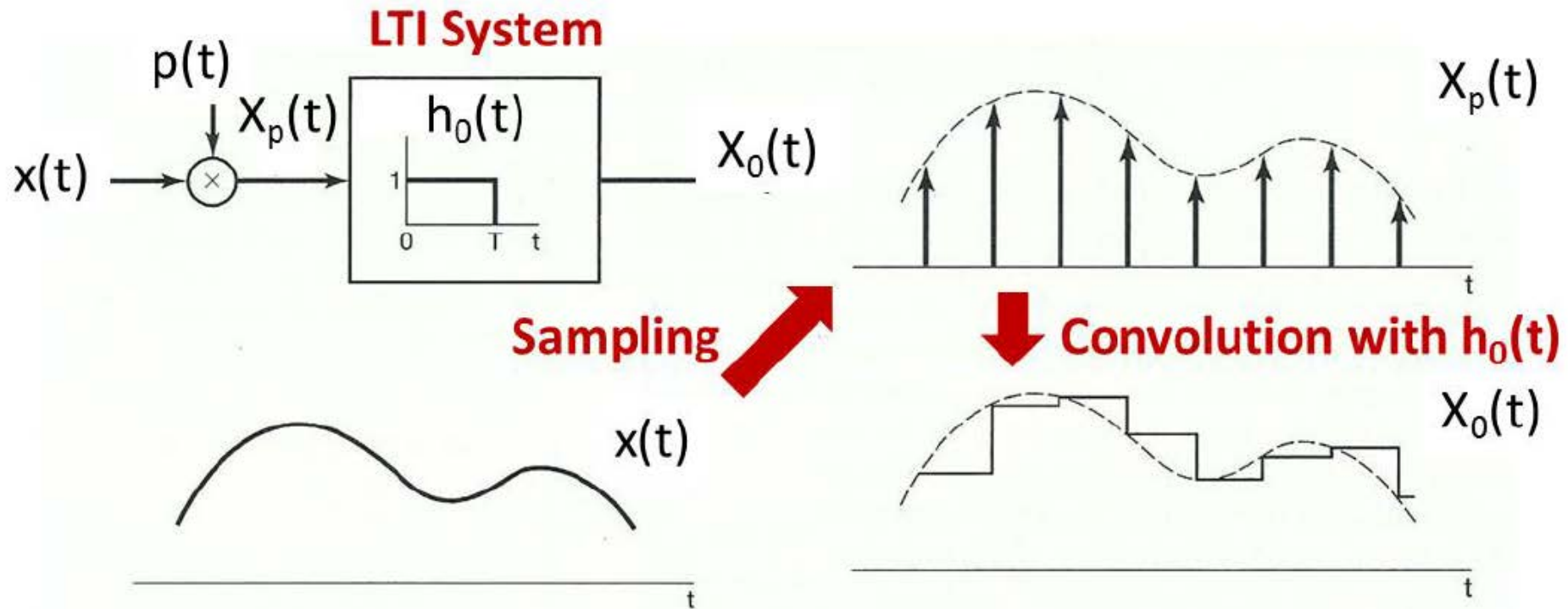
$$x_p(t) = \sum_{n=-\infty}^{\infty} x(nT)\delta(t - nT)$$

- Alternative approach: zero-order hold



- How to interpret the system of "zero-order hold" mathematically?

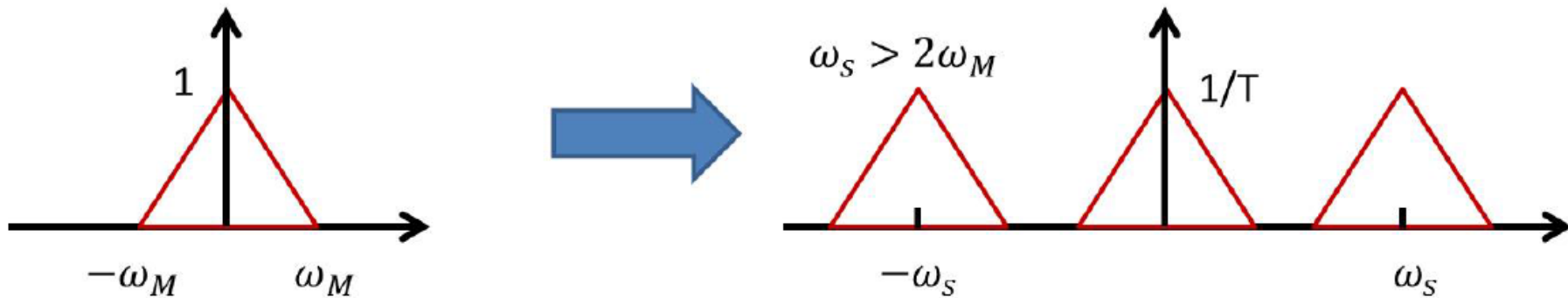
Interpretation of zero-order hold



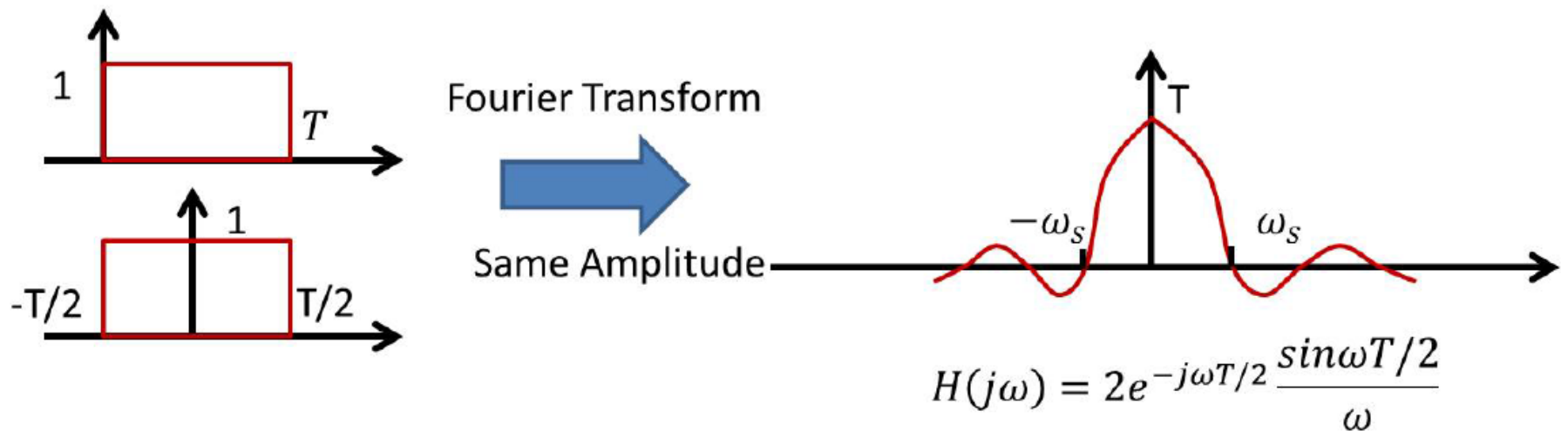
- Zero-order hold: sampling + interpolation with rectangular impulse response
- An approximation of the signal to be sampled.

Frequency analysis (1/2)

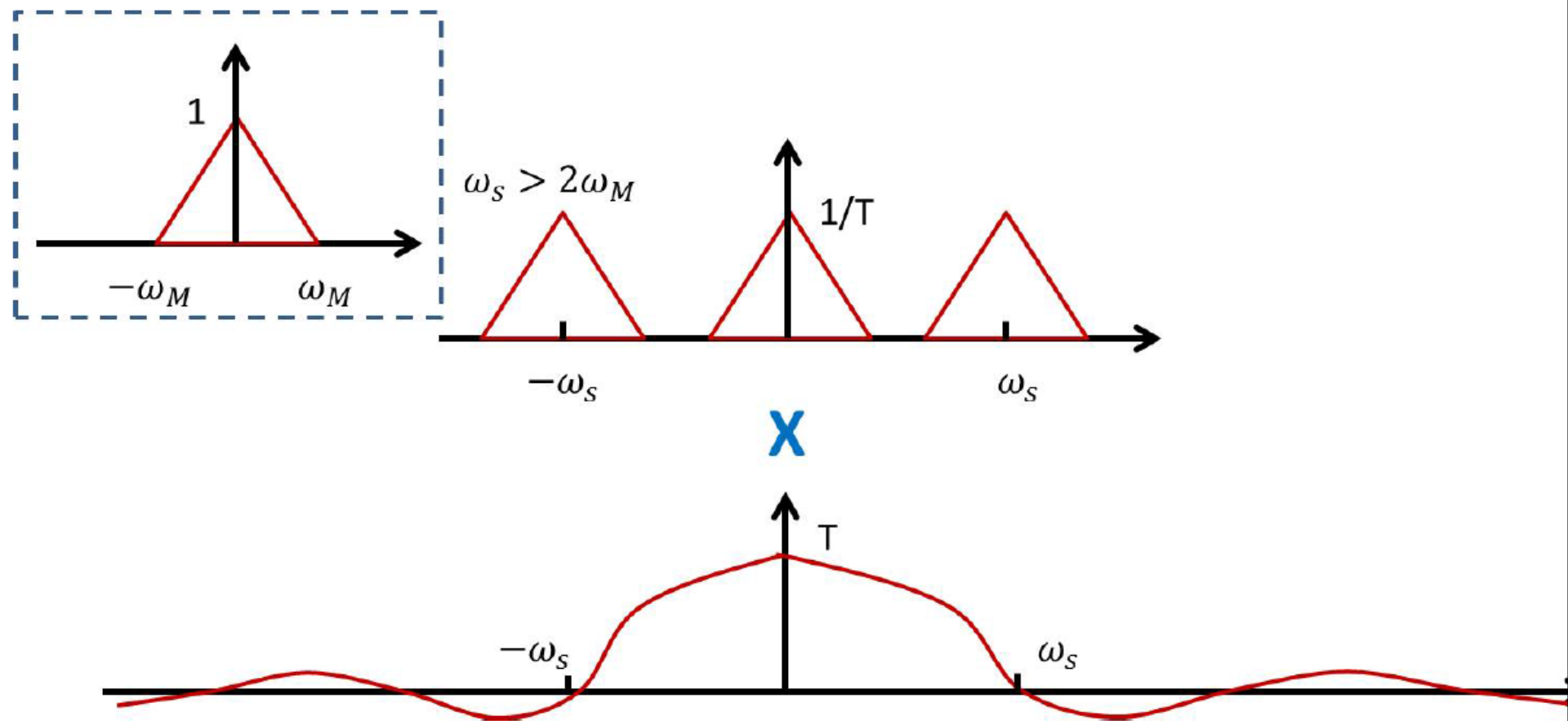
- Step 1: Impulse-train sampling



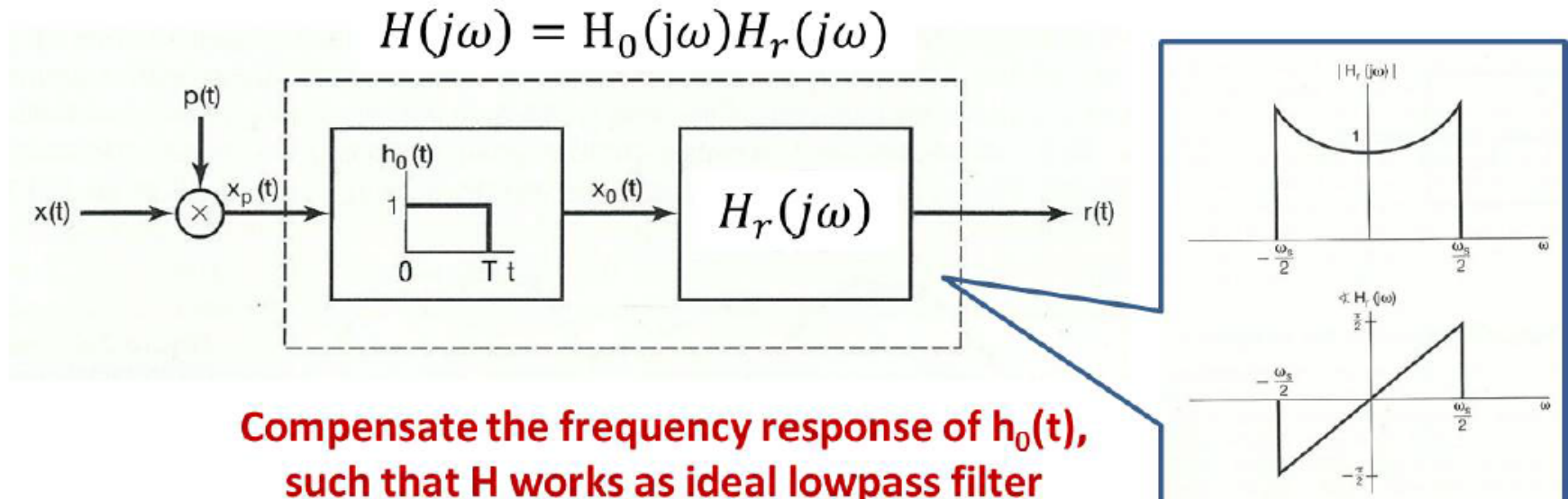
- Step 2: Frequency response of $h_0(t)$



Frequency analysis (2/2)



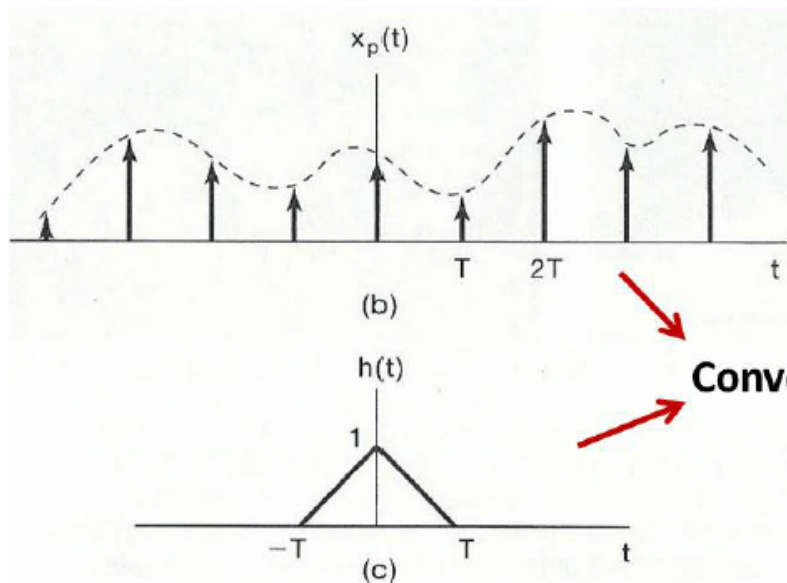
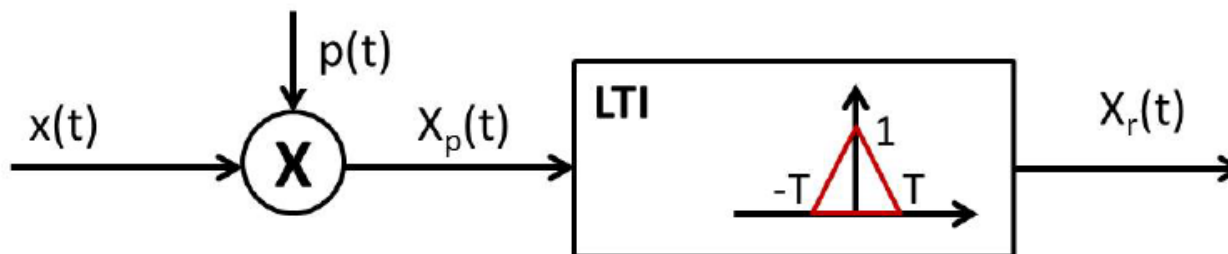
Reconstruction



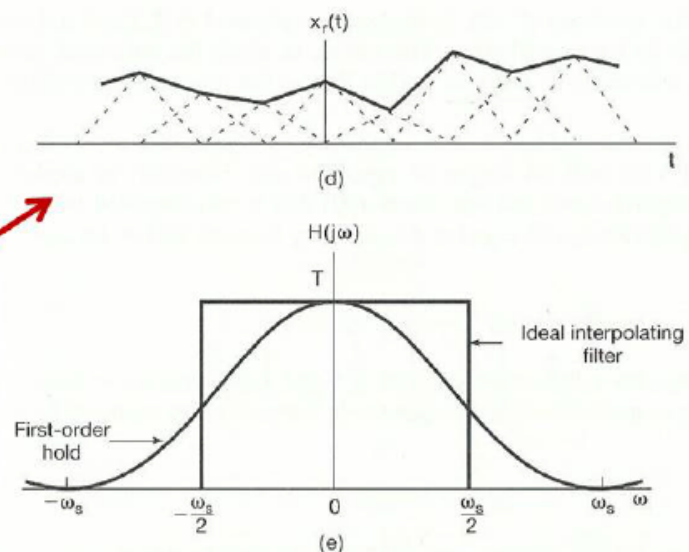
$$H_r(j\omega) = e^{j\omega T/2} \frac{\omega T}{2 \sin \omega T/2} \quad -\frac{\omega_s}{2} \leq \omega \leq \frac{\omega_s}{2}$$

- $H(j\omega)$ should be an ideal low-pass filter from $-\omega_s/2$ to $\omega_s/2$

First-order hold



Convolution



- First-order hold: sampling + interpolation with triangular wave
- How to reconstruct?

Summary: Sampling approaches

